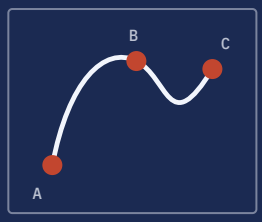
OPEN-SOURCE RESEARCH PLATFORM · BAR-ILAN UNIVERSITY

ILT: spotting literacy difficulties early, from how a child learns to write a new letter

The Invented Letter Task (ILT) Platform turns a validated paper-and-digitizer research test into a digital system that schools, research labs and clinics can run on pen tablets, free of licence fees.



THE NEED

Children who struggle are found too late

Dyslexia, dysgraphia and reading-comprehension difficulties are usually identified only after a child has already fallen behind. Children from low socio-economic backgrounds are at the highest risk of going unnoticed.

Research by Dr. Hagit Hollander and Prof. Esther Adi-Japha shows that how well a second-grader learns and retains a new handwriting movement predicts their reading and writing at the end of the school year. A test of that learning can flag children at risk months before formal failure.

HOW IT WORKS

A child writes one invented letter, again and again

On a pen tablet, the child writes a letter they have never seen: one continuous stroke through three dots, in a grid of 15 boxes. Because the letter is new, the test measures pure motor learning, not prior knowledge or home language.

Day 1	Day 2	Day 14
Learning: speed and accuracy across practice blocks	Overnight consolidation of the new skill	Retention without further practice

An evaluator follows each stroke live on a laptop. Each block is scored automatically when it ends, and every session can be replayed stroke by stroke.

BUILT FOR RESEARCH

Raw data first, privacy by design

- Every pen sample is kept, so scores can be recomputed when the scoring method improves.
- No names or identity data are stored. Analysts see only de-identified research data.
- Hebrew and English with full right-to-left support; new languages need no code changes.
- Fully open source, built only on open-source components.

WHAT IT CAPTURES

Measured on the Wacom tablet

240 Hz	pen samples per second while writing
~8,000	pressure steps, plus pen tilt
10 mm	hover range: time with the pen in the air is measured
< 15 ms	target delay of the evaluator's live view
500	concurrent sessions in the cloud design

STATUS · NOVEMBER 2026

Core system working; pilot next

- [x] Foundations: database, scoring engine, Hebrew and English apps
- [x] Pen capture, tablet pairing and live mirror, tested on the target tablet
- [x] Full Day 1 / 2 / 14 sessions with scoring, summary and replay
- [ ] Administration, research portal and analytics
- [ ] Intervention program with real letters
- [ ] Pilot, 2026/27 school year: 50 children on tablets, 50 on paper, plus 50 earlier datasets

HOW YOU CAN HELP

Sponsorship opportunities

- **Pen tablets** for field kits that test one child, or four children at once.
- **Cloud hosting** for the central research system: about 7 virtual servers at full scale.
- **Research and development funding** to complete the platform and run the pilot.
- **School and clinic partners** to take part in the pilot.

Sponsors are acknowledged in publications, conference talks and the project's open-source repository.

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With Dr. Hagit Hollander & Prof. Esther Adi-Japha, School of Education, Bar-Ilan University

Research basis: Skill learning as a predictor of literacy abilities and related impairments among second-graders from low-SES, *Learning and Individual Differences*, 2025.